Chapter 5. Statistical Inference

Estimation and hypothesis testing

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5.1 Estimation

Motivation. Statistical inference draws conclusions about a population from a sample and attaches a probability-based measure of reliability to them. Its two basic forms are estimation and significance testing; this section explains estimation through the population mean.

The usual targets of inference are population characteristics such as the mean, standard deviation, proportion or correlation. Computing an estimate of such a characteristic from a sample *together with its margin of error* is called **estimation**.

5.1.1 Estimating a population mean

Illustration 5-1: tire lifetimes

A tire maker develops a new process expected to raise the mean tire life, while the standard deviation stays at the old value $\sigma = 5000$ km. A trial of $n = 100$ tires gives $\bar{x} = 38000$ km. The natural estimate of the mean life μ is $\hat{\mu} = \bar{x} = 38000$ km.

How reliable is this method? By the CLT, $\bar{X}$ is approximately $N(\mu, \sigma^2/100)$ with standard deviation $\sigma/\sqrt{n} = 5000/10 = 500$ km. Hence

(a) with probability about 0.95, $\bar{X}$ lies within $1.96\,\sigma/\sqrt{n} = 980$ of μ;

(b) the event $|\bar{X} - \mu| < 980$ is the same as $\mu \in (\bar{X} - 980, \bar{X} + 980)$;

(c) if the experiment were repeated indefinitely, about 95% of the intervals $(\bar{x} - 980, \bar{x} + 980)$ would contain μ.

For the observed sample this interval is $(37020, 38980)$. Either μ lies in it, or the sample is one of the unusual 5% for which $\bar{x}$ is more than 980 from μ. We cannot tell which, but the *method* is right 95 times in 100, so we state $37020 < \mu < 38980$ with **confidence level** 95%; the interval is a 95% **confidence interval**. In short, $\hat{\mu} = \bar{x} = 38000$ with $|\mu - \hat{\mu}| \le 1.96\,\sigma/\sqrt{n} = 980$.

Definition (Estimate, estimator, standard error)

The number $\hat{\mu} = \bar{x}$ is an **estimate** of μ, and $1.96\,\sigma/\sqrt{n}$ is its 95% **margin of error**. The statistic $\hat{\mu} = \bar{X}$ used for estimation is an **estimator** (the estimator relates to the estimate as a random variable to its observed value). The standard deviation of an estimator is its **standard error**:

$$\mathrm{SE}(\hat{\mu}) = \sqrt{\mathrm{Var}(\hat{\mu})} = \sqrt{\mathrm{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}.$$

For large n,

$$P\Big(-z_{\alpha/2} \le \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\Big) \approx P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1 - \alpha.$$

Definition (Estimating μ when σ is known)

(a) Estimator and standard error: $\hat{\mu} = \bar{X}$, $\mathrm{SE}(\hat{\mu}) = \sigma/\sqrt{n}$.
(b) $100(1-\alpha)\%$ margin of error: $z_{\alpha/2}\,\mathrm{SE}(\hat{\mu}) = z_{\alpha/2}\,\sigma/\sqrt{n}$.
(c) $100(1-\alpha)\%$ confidence interval: $\Big(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\Big)$.

These statements are exact for a normal population and approximate (large n) otherwise. Common confidence levels are 90%, 95%, 99%, with $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.005} = 2.576$.

Illustration 5-2

From a normal population with $\sigma = 10$, a sample of $n = 25$ gives $\bar{x} = 40$. Then $\hat{\mu} = 40$, the 90% margin of error is $1.645 \cdot 10/\sqrt{25} = 3.29$, and the 90% confidence interval is (36.71, 43.29).

Meaning of the confidence level. In the setting of Illustration 5-2, twenty samples of size 25 give twenty intervals $\bar{x} \pm 3.29$ with different centres; in the book's simulation 18 of the 20 contain μ. A 90% confidence level means that, over repeated sampling, 90% of such intervals contain μ.

Example 5-1 (Cholesterol of low-income residents)

A survey of $n = 49$ low-income Central American residents found mean blood cholesterol $\bar{x} = 157.02$ mg/L. Assume cholesterol is approximately normal with $\sigma = 30$ mg/L. Find a 95% confidence interval for μ.
Solution. $\bar{x} \pm 1.96\,\sigma/\sqrt{n} = 157.02 \pm 1.96 \cdot 30/7 = 157.02 \pm 8.40$, i.e. (148.62, 165.42) mg/L.

At 99% the interval is $\bar{x} \pm 2.576\,\sigma/\sqrt{n} = (145.98, 168.06)$: higher confidence gives a longer interval, and excessively high confidence makes the interval uselessly wide. The length of a $100(1-\alpha)\%$ interval is

$$\left(\bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) - \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 2z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

twice the margin of error, so it can be controlled through n.

Definition (Sample size for estimating μ (σ known))

To make the $100(1-\alpha)\%$ margin of error at most d (interval length at most $2d$), take

$$n \geq \left(z_{\alpha/2}\frac{\sigma}{d}\right)^2.$$

(Indeed $z_{\alpha/2}\sigma/\sqrt{n} \leq d \iff \sqrt{n} \geq z_{\alpha/2}\sigma/d$.)

Example 5-2 (Required sample size)

In Example 5-1, how many residents must be surveyed for the 95% margin of error to be at most 5 mg/L?
Solution. With $\sigma = 30$, $n \geq (1.96 \cdot 30/5)^2 = 138.2976$, so at least 139 residents are needed.

5.1.2 Parameters, point and interval estimation

In practice σ is rarely known; the general case is treated in Chapter 6. A numerical summary of a population, such as μ or σ, is a **parameter**, generically θ. An estimator of θ based on $X_1, \ldots, X_n$ is written $\hat{\theta}(X_1, \ldots, X_n)$, or simply $\hat{\theta}$ (also for the estimate).

Definition (Confidence interval for a parameter)

If statistics $L = L(X_1, \ldots, X_n)$ and $U = U(X_1, \ldots, X_n)$ satisfy

$$P(L < \theta < U) = 1 - \alpha,$$

then (L, U) is a $100(1-\alpha)\%$ confidence interval for θ. The confidence level has the same repeated-sampling meaning as before.

Illustration 5-3: estimating σ and σ^2

The usual estimator of σ is the sample standard deviation, and that of σ^2 the sample variance:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2} = S, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 = S^2.$$

Giving only an estimate of the parameter is **point estimation**; giving a confidence interval is **interval estimation**. Together they constitute **estimation of parameters**.

Example 5-3 (Thickness of plastic sheets)

The variability of thickness is a key quality characteristic. Ten sheets measured (mm):

226	228	226	225	232	228	227	229	225	230

Estimate the population standard deviation σ.
Solution. $\bar{x} = \frac{1}{10}(226 + 228 + \cdots + 230) = 227.6$ mm and

$$s^2 = \frac{1}{10-1}\{(226-227.6)^2 + (228-227.6)^2 + \cdots + (230-227.6)^2\} = 5.1556,$$

so $\hat{\sigma} = s = \sqrt{5.1556} = 2.27$ mm.

Other estimators exist (e.g. the trimmed mean for μ, more stable when there are outliers), but this book uses $\bar{X}$ and S.

5.2 Significance tests and hypothesis tests

Motivation. Sometimes the question is not "how large is μ?" but "do the data give clear evidence against a stated claim?". Significance testing measures the strength of such evidence by a probability, the P-value.

5.2.1 Null and alternative hypotheses

Illustration 5-4: testing Mendel's law

A scientist crossed genetically modified peas and observed in the second generation 860 round-yellow, 340 round-green, 240 wrinkled-yellow and 160 wrinkled-green peas. Mendel's theory predicts the ratio $9:3:3:1$. Do the data strongly suggest a new theory?

Science advances by finding strong evidence against established theories. Here the claim under scrutiny is the **hypothesis** "the four types occur in ratio $9:3:3:1$".

Definition (Null and alternative hypotheses; significance test)

A hypothesis set up in order to look for evidence *against* it is the **null hypothesis** H_0; the hypothesis proposed as the alternative to H_0 is the **alternative hypothesis** H_1. A **significance test** evaluates the strength of the evidence against H_0. Its logic resembles proof by contradiction: if H_0 were true, the observed data would be very unlikely.

Illustration 5-5: sprinkler temperature

Sprinklers are designed to activate at 55°C. The process is abnormal if the mean activation temperature μ satisfies $\mu < 55$ or $\mu > 55$. Since we seek clear evidence of abnormality,

$$H_0 : \mu = 55, \qquad H_1 : \mu \neq 55.$$

Illustration 5-6: paint drying time

A paint dries in 75 minutes on average ($\sigma = 9.4$). A lab claims a new additive shortens the drying time, and the company will adopt it only on clear evidence. With μ the mean drying time with the additive,

$$H_0 : \mu = 75, \qquad H_1 : \mu < 75.$$

An alternative containing values on both sides of the reference value (as $\mu \neq 55$) is **two-sided**; one containing one side only (as $\mu < 75$) is **one-sided**. The claim we want to establish from the data goes into H_1.

Choice of the null value

With $H_1 : \mu < 75$ the logical null is its negation $H_0 : \mu \geq 75$. Using instead the boundary case $H_0 : \mu = 75$, the value hardest to distinguish from H_1, does not change the testing procedure, so the null is stated at the boundary value.

Evidence for H_1 is evaluated through an estimator of the parameter: the farther the estimate lies from the value specified by H_0 in the direction of H_1, the stronger the evidence.

Illustration 5-7

For $H_0 : \mu = 55$, $H_1 : \mu \neq 55$, the evidence is sought in $\bar{X}$; since H_1 is two-sided, $\bar{X}$ far from 55 in *either* direction supports H_1.

Illustration 5-8

For $H_0 : \mu = 75$, $H_1 : \mu < 75$ (left-sided), values of $\bar{X}$ far *below* 75 support H_1.

A statistic such as $\bar{X}$ used in a test is a **test statistic**.

5.2.2 The P-value

Illustration 5-9: P-value for the sprinklers

For $H_0 : \mu = 55$, $H_1 : \mu \neq 55$, $n = 9$ sprinklers gave $\bar{x} = 55.63$°C; assume normality with $\sigma = 0.9$. The observed deviation is $\bar{x} - 55 = 0.63$. The probability, under H_0, of evidence at least as strong as observed is

$$P\Big(\Big|\frac{\bar{X} - 55}{\sigma/\sqrt{n}}\Big| \geq 2.1 \;\Big|\; \mu = 55\Big) = P(|Z| \geq 2.1) = 0.0357,$$

since $0.63/(0.9/3) = 2.1$. Repeating the survey, stronger evidence against H_0 would arise fewer than 4 times in 100 if H_0 were true; the data strongly suggest $\mu \neq 55$.

Definition 5-1 (Significance probability (P-value))

The **P-value** is the probability, computed under H_0, that the test statistic takes a value at least as extreme as the observed value in the direction supporting H_1. The smaller the P-value, the stronger the evidence against H_0.

Illustration 5-10: P-value for the paint

For $H_0 : \mu = 75$, $H_1 : \mu < 75$, $n = 100$ samples with the additive gave $\bar{x} = 73.5$ ($\sigma = 9.4$). By the CLT, $\bar{X} \approx N(\mu, 9.4^2/100)$, so

$$P = P(\bar{X} \le 73.5 \mid \mu = 75) = P\Big(\frac{\bar{X} - 75}{9.4/\sqrt{100}} \le -1.5957 \,\Big|\, \mu = 75\Big) \approx P(Z \le -1.5957) = 0.0553.$$

The data fairly strongly suggest a shorter drying time; the strength of evidence is about 5.5%.

5.2.3 Significance level

"Strong" is relative, so a reference value α, the **significance level**, is often fixed in advance and the P-value compared with it. Usually $\alpha = 0.05$ or 0.01. With $\alpha = 0.05$, a P-value ≤ 0.05 is reported as "statistically significant ($P \le 0.05$)".

Definition 5-2 (Statistical significance)

If the P-value is at most the significance level α, the result is **statistically significant at level α**; the evidence against H_0 is stronger than the prescribed level.

Definition (Procedure of a significance test)

(a) State H_0 and H_1.
(b) [Optional] Fix the significance level α.
(c) Compute the observed value of the test statistic (based on an estimator of the relevant parameter).
(d) Compute the P-value from the sampling distribution of the test statistic under H_0; it measures the evidence against H_0.
(e) [Optional] Compare the P-value with α: if $P \le \alpha$ the result is significant.

Steps (c) and (d) are mechanical (tables or software), but H_0, H_1 and the interpretation are the investigator's responsibility.

5.2.4 Tests for μ with σ known

For $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ the standardized statistic and its observed value are

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \qquad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}},$$

and $P = P(\bar{X} \le \bar{x} \mid \mu = \mu_0) \approx P(Z \le z)$ (in Illustration 5-10, $z = -1.5957$ and $P = 0.0553$). For $H_1 : \mu > \mu_0$, $P \approx P(Z \ge z)$; for $H_1 : \mu \ne \mu_0$, deviations of $\bar{X}$ from μ_0 in both directions count, so

$$P(\bar{X} - \mu_0 \le -|\bar{x} - \mu_0| \text{ or } \bar{X} - \mu_0 \ge |\bar{x} - \mu_0| \mid \mu = \mu_0) \approx P(|Z| \ge |z|).$$

Definition (Significance test for μ (σ known))

For $H_0 : \mu = \mu_0$ use $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$. With observed value z, the P-value is
(a) $P = P(Z \geq z)$ if $H_1 : \mu > \mu_0$;
(b) $P = P(Z \leq z)$ if $H_1 : \mu < \mu_0$;
(c) $P = P(|Z| \geq |z|)$ if $H_1 : \mu \neq \mu_0$.

The P-value is exact for a normal population and approximate for large n otherwise.

Example 5-4 (New tire process)

Current tires last on average 37,000 km with standard deviation 5,000 km. A trial of 100 tires from a new process gives $\bar{x} = 38000$ km. Assuming $\sigma = 5000$ is unchanged, do the data show that the new process is successful?
Solution. We want to establish $\mu > 37000$: $H_0 : \mu = 37000$, $H_1 : \mu > 37000$. The observed statistic is

$$z = \frac{38000 - 37000}{5000/\sqrt{100}} = 2.$$

Z is approximately standard normal ($n = 100$), and H_1 is right-sided, so $P = P(Z \geq 2) = 1 - 0.9772 = 0.0228$. The data strongly suggest that the new process increases tire life.

Example 5-5 (Concentration of an active ingredient)

A drug should contain 0.85 g/L of an active ingredient; repeated measurements on one specimen are $N(\mu, 0.0123^2)$. Three measurements of a specimen from a new method average $\bar{x} = 0.8404$ g/L. Is there evidence that $\mu \neq 0.85$?
Solution. $H_0 : \mu = 0.85$, $H_1 : \mu \neq 0.85$, and

$$z = \frac{0.8404 - 0.85}{0.0123/\sqrt{3}} = -1.35.$$

The population is normal, so $Z \sim N(0,1)$ exactly, and $P = P(|Z| \geq 1.35) = 2P(Z \leq -1.35) = 2 \times 0.0885 = 0.177$. A P-value above 17% gives no clear evidence that the true concentration differs from 0.85 g/L.

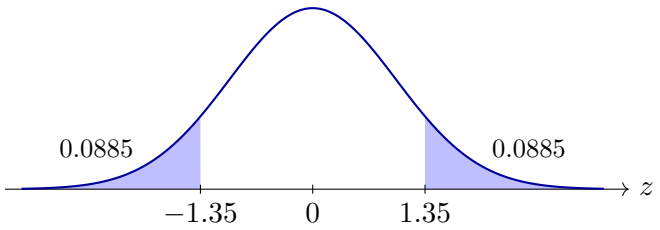


Figure 1: Two-sided P-value in Example 5-5: $P = P(|Z| \geq 1.35) = 0.177$.

5.2.5 Rejection regions and hypothesis testing

If the required strength of evidence α is fixed beforehand, the P-value need not be computed.

Illustration 5-11

In Example 5-4 ($H_0 : \mu = 37000$, $H_1 : \mu > 37000$) take $\alpha = 0.05$. We must decide whether $P(Z \geq z) \leq 0.05$. Two equivalent ways:
(a) compute the P-value from $z = 2$ as in Example 5-4;
(b) compare z with the upper 5% point $z_{0.05} = 1.645$. Since $z = 2 > 1.645$, i.e. $z \geq 1.645$ holds, the result is significant at 5%. Had $z < 1.645$, it would not be.

Definition (Rejection region; hypothesis test)

For a fixed level α, the set of values of the test statistic for which significance at level α is declared (e.g. $z \geq 1.645$) is the **rejection region** of H_0. The conclusion is stated as "H_0 is rejected at level α" or "H_0 cannot be rejected at level α". Deciding in this way is called **hypothesis testing**:
(a) find the rejection region corresponding to α;
(b) compute the observed value of the test statistic;
(c) check whether it lies in the rejection region.

Definition (Hypothesis test for μ (σ known, level α))

For $H_0 : \mu = \mu_0$ with $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ and observed value z, the rejection region is
(a) $z \geq z_\alpha$ if $H_1 : \mu > \mu_0$;
(b) $z \leq -z_\alpha$ if $H_1 : \mu < \mu_0$;
(c) $|z| \geq z_{\alpha/2}$ if $H_1 : \mu \neq \mu_0$.

Frequently $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.01} = 2.326$, $z_{0.005} = 2.576$.

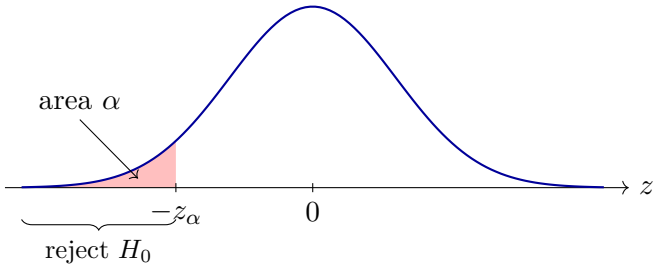


Figure 2: Rejection region $z \leq -z_\alpha$ for $H_1 : \mu < \mu_0$.

Example 5-6 (A faster pain reliever)

An existing painkiller takes on average 30 minutes to act ($\sigma = 5$). For a new painkiller, 50 randomly chosen patients gave $\bar{x} = 28.5$ minutes; assume normality with $\sigma = 5$. Test at level 5% whether the new drug acts faster.
Solution. $H_0 : \mu = 30$, $H_1 : \mu < 30$. The rejection region at 5% is

$$z = \frac{\bar{x} - 30}{5/\sqrt{50}} \leq -z_{0.05} = -1.645.$$

Here $z = \frac{28.5-30}{5/\sqrt{50}} = -2.12$ lies in the rejection region, so H_0 is rejected at 5%: the evidence that the new drug is faster exceeds the prescribed level.

Example 5-7 (Sprinklers at level 1%)

In Illustrations 5-5 and 5-9 ($n = 9$, $\bar{x} = 55.63$, $\sigma = 0.9$), test $H_0 : \mu = 55$ against $H_1 : \mu \neq 55$ at level 1%.
Solution. $z = \frac{55.63-55}{0.9/\sqrt{9}} = 2.1$, and the rejection region is $|z| \geq z_{0.005} = 2.576$. Since $2.1 < 2.576$, H_0 cannot be rejected at 1%: the evidence of a process abnormality is not stronger than the 1% level.

The conclusion depends on α. In Example 5-6, $P = P(Z \leq -2.12) = 0.0170$: rejected at $\alpha = 0.05$ but not at $\alpha = 0.01$. In Example 5-7, $P = P(|Z| \geq 2.1) = 0.0357$: not rejected at 0.01 but rejected at 0.05. The P-value reports the evidence without fixing α; when a decision is required, α must be chosen with the practical context in mind.

5.2.6 Two kinds of error

A level-α test is a rule for choosing between H_0 and H_1, and it can err in two ways.

Decision \ Truth	H_0 true	H_1 true
Do not reject H_0 (choose H_0)	correct	Type II error
Reject H_0 (choose H_1)	Type I error	correct

Illustration 5-12: Type I error probability

In Example 5-6 the level-α rejection region is $\frac{\bar{X}-30}{5/\sqrt{50}} \leq -z_\alpha$, so

$$P(\text{Type I error}) = P\Big(\frac{\bar{X}-30}{5/\sqrt{50}} \leq -z_\alpha \;\Big|\; \mu = 30\Big) = P(Z \leq -z_\alpha) = \alpha.$$

In general a test with the level-α rejection region has Type I error probability at most α (with $H_0 : \mu \geq \mu_0$, the probability is largest at the boundary $\mu = \mu_0$). Fixing α thus sets the tolerated Type I error rate.

Illustration 5-13: Type II error probability

In Example 5-6, a true mean of 28 minutes would be a notable improvement. If $\mu = 28$, a Type II error occurs when H_0 is not rejected:

$$\begin{aligned} P\Big(\frac{\bar{X}-30}{5/\sqrt{50}} > -1.645 \;\Big|\; \mu = 28\Big) &= P\Big(\frac{\bar{X}-28}{5/\sqrt{50}} + \frac{28-30}{5/\sqrt{50}} > -1.645 \;\Big|\; \mu = 28\Big) \\ &= P(Z > 2\sqrt{2} - 1.645) = P(Z > 1.183) \approx 0.1190. \end{aligned}$$

Just as α bounds the Type I error, one may require the Type II error probability at a specified alternative to be at most β; this is achieved by choosing n.

Example 5-8 (Sample size for given α and β)

In Example 5-6, find n such that the level-0.05 test has Type II error probability at most $\beta = 0.1$ when $\mu = 28$.
Solution. As in Illustration 5-13,

$$P\Big(\frac{\bar{X}-30}{5/\sqrt{n}} > -1.645 \;\Big|\; \mu = 28\Big) = P\Big(\frac{\bar{X}-28}{5/\sqrt{n}} + \frac{28-30}{5/\sqrt{n}} > -1.645 \;\Big|\; \mu = 28\Big)$$

$$= P\Big(Z > \frac{30-28}{5/\sqrt{n}} - 1.645\Big) \le 0.1.$$

Since $z_{0.1} = 1.282$, this requires $\frac{30-28}{5/\sqrt{n}} - 1.645 \ge 1.282$, i.e.

$$n \ge \Big\{\frac{1.645 + 1.282}{(30-28)/5}\Big\}^2 = 53.546.$$

The required sample size is $n = 54$.

Definition (Sampling inspection; power)

Choosing n and the rejection region so that the Type I error probability is at most α and the Type II error probability at $\mu = \mu_1$ is at most β is called **sampling inspection** (widely used in statistical quality control). The probability of *not* making a Type II error when $\mu = \mu_1$, i.e. of rejecting H_0, is the **power** at $\mu = \mu_1$.

Illustration 5-14: power

For the 5% test of Illustration 5-13, the power at $\mu = 28$ is

$$P\Big(\frac{\bar{X} - 30}{5/\sqrt{50}} \le -1.645 \;\Big|\; \mu = 28\Big) = 1 - 0.1190 = 0.8810.$$

Conditions for valid inference

For the methods of this chapter to be valid:
(a) the sample must be drawn at random;
(b) the assumptions on the population distribution must be checked.